

Sample Exam

Medical Image Analysis

Student name:_____

Immatriculation number:_____

Exam Questions

Note: All the questions have equal weight.

1. In 2D acquisitions, slices are acquired in one pre-defined plane and stacked along the remaining direction: axial slices stack head-to-toe, sagittal slices left-to-right, coronal slices anterior-to-posterior. Voxel sizes are given as in-plane (two values) and through-plane (the slice thickness), and the field-of-view is image size $\times$ voxel size.

You are handed a volume with the following properties, but the acquisition plane is not recorded: field-of-view 240 mm head-to-toe, 240 mm left-to-right, 180 mm anterior-to-posterior; voxel size in-plane $0.75 \times 0.75 \text{ mm}^2$, through-plane 4.5 mm.

- (a) In which plane was this volume acquired? Justify your answer from the numbers.

Solution: Answer: a **coronal** acquisition.

The through-plane direction is the one whose extent is sampled at 4.5 mm, and only anterior-to-posterior works: $180/4.5 = 40$ slices, an integer. If either of the other two directions were through-plane, it would need $240/4.5 = 53.3$ slices — not an integer, so the numbers are inconsistent with any other assignment. Slices stacked along anterior-to-posterior lie in the coronal plane.

- (b) How many voxels does the volume have in the head-to-toe, left-to-right and anterior-to-posterior directions, respectively?

Solution: Answer: $320 \times 320 \times 40$ voxels.

Head-to-toe and left-to-right are in-plane: $240/0.75 = 320$ each. Anterior-to-posterior is through-plane: $180/4.5 = 40$.

- (c) The volume is resampled to isotropic $1.5 \times 1.5 \times 1.5 \text{ mm}^3$ voxels. Give the resize factors and the resulting image size, and verify that the field-of-view is unchanged.

Solution: Answer: resize factors $0.5 \times 0.5 \times 3.0$, resulting image size $160 \times 160 \times 120$ voxels.

The factor is the ratio of old to new voxel size: $0.75/1.5 = 0.5$ in-plane, $4.5/1.5 = 3.0$ through-plane. Sizes: $320 \cdot 0.5 = 160$, $320 \cdot 0.5 = 160$, $40 \cdot 3.0 = 120$. Field-of-view check: $160 \cdot 1.5 = 240$, $160 \cdot 1.5 = 240$, $120 \cdot 1.5 = 180 \text{ mm}$ — unchanged, as it must be. (The

anterior-to-posterior direction is upsampled by 3: two of every three samples there are interpolated, not measured.)

2. In bilinear interpolation the intensity at a query point x_1 (with $x_1^0 \leq x_1 \leq x_1^1$) is first interpolated along one dimension from the two neighbouring pixels. Which of the two expressions below is the correct linear interpolation? Please circle the correct choice.

$$I(x_1) = \frac{x_1^1 - x_1}{x_1^1 - x_1^0} I(x_1^0) + \frac{x_1 - x_1^0}{x_1^1 - x_1^0} I(x_1^1) \qquad I(x_1) = \frac{x_1 - x_1^0}{x_1^1 - x_1^0} I(x_1^0) + \frac{x_1^1 - x_1}{x_1^1 - x_1^0} I(x_1^1)$$

Solution: Answer: Left one.

Each pixel's weight must be proportional to the query point's distance from the *other* pixel — nearness earns weight. The sanity check takes two seconds: set $x_1 = x_1^0$. The left expression gives $1 \cdot I(x_1^0) + 0 \cdot I(x_1^1) = I(x_1^0)$, as an interpolant must. The right expression gives $I(x_1^1)$ — the wrong pixel entirely; its weights are inverted. Bilinear interpolation applies this step along x_1 and then along x_2 (trilinear adds the third dimension, using the 8 cube corners).

3. Probabilistic reasoning underlies most of this course.
- (a) If each pixel intensity is treated as a realization of a random variable, the (normalized) histogram and the cumulative histogram of an image approximate two standard probabilistic objects. Which two — and what assumption about the pixels makes this approximation valid?

Solution: Answer: the normalized histogram approximates the **probability density function** of the intensity, and the cumulative histogram approximates the **cumulative distribution function**.

The assumption is that each pixel intensity is an **independent realization** of the same random variable I — only then is counting pixels per intensity bin an estimate of the underlying distribution.

- (b) The Maximum-A-Posteriori estimate is $\arg \max_l p(i | l) p(l)$ and the Maximum Likelihood estimate is $\arg \max_l p(i | l)$. Under what condition on the prior are the two identical?

Solution: Answer: when the prior is **uniform**, $p(l) = c$ for all l .

A constant factor does not change the location of a maximum, so the prior drops out of the $\arg \max$ and MAP reduces to MLE. Conversely, whenever anyone uses MLE they are implicitly assuming that no value of l is more plausible than any other before seeing the data.

4. Non-local means computes $\hat{I}(x) = \sum_y w(x, y) J(y)$ with weights from the similarity of the patches $N(x)$ and $N(y)$, a degree-of-filtering parameter h , and averaging restricted to a search window $W(x)$. Please indicate whether the statement is true or false by circling the correct choice in each of the following.

- (a) True **FALSE** Choosing the patch to be the entire image, $N(x) = \Omega$, reduces non-local means to Gaussian filtering.
- (b) **TRUE** False Choosing the patch to be the single pixel, $N(x) = \{x\}$, reduces non-local means to Yaroslavsky's filter, a special case of bilateral filtering.
- (c) True **FALSE** Increasing h decreases the amount of smoothing.
- (d) **TRUE** False The averaging is restricted to a search window $W(x)$ rather than the whole image mainly to reduce the computational cost.

Solution: (a) FALSE — it reduces to *mean* filtering, not Gaussian filtering. If every pixel's patch is the whole image, all patches are identical, every weight is equal, and the weighted average degenerates to the plain average.

(b) TRUE — with a single-pixel “patch” the weight compares raw grey values, which is exactly the neighbourhood-filter family (Yaroslavsky, bilateral). This is also why NLM is more robust: a single noisy value is a fragile basis for similarity, a whole patch is not.

(c) FALSE — h is the degree of filtering: larger h makes the exponential weights decay more slowly, so more (less similar) patches contribute and the result is *smoother*. h should be tied to the noise level, which is worth estimating first.

(d) TRUE — and it also encodes a mild prior that useful matches are nearby. The three parameters of the method are exactly h , the patch size, and the search-window size.

5. A 2D landmark-based registration uses thin-plate splines with the $N = 50$ landmarks as control points, combined with a global affine part. How many parameters does the transformation have? How does this compare with a naive deformable parameterization of a 256×256 image that stores one independent displacement vector per pixel?

Solution: Answer: the TPS model has $50 \times 2 + 6 = \mathbf{106}$ parameters; the naive field has $256 \times 256 \times 2 = \mathbf{131\,072}$ — about 1 240 times more.

Each control point carries a coefficient vector $d_n \in \mathbb{R}^2$ (100 parameters), and the 2D affine part adds $4 + 2 = 6$. The comparison is the point of the question: both are “deformable” models, but the TPS parameterization builds smoothness in and needs three orders of magnitude fewer parameters, while the naive field is not only huge but can produce noisy, topology-breaking transformations for generic parameter values.

6. An Active Appearance Model extends a statistical shape model with a model of texture. Please describe briefly (i) how the texture model is built from the training images, and (ii) how shape and texture are combined into a single set of appearance parameters. You do not need to write down the equations.

Solution: Answer: (i) *Building the texture model*. Every training image is **warped to the mean shape**, which removes the shape variation and yields a shape-free patch. The intensities are sampled from this shape-normalized image into a texture vector and **normalized**

in intensity to eliminate lighting variation. PCA is then applied to the normalized texture vectors, exactly as it was applied to the shape coordinates: $g \approx \bar{g} + P_g b_g$.

(ii) *Combining shape and texture.* Each training sample is now summarized by two parameter vectors, b_s for shape and b_g for texture — and the two are correlated (a shape change usually comes with a texture change). They are stacked into a single vector, with a weighting W_s on the shape part to account for the different units of shape and texture, and a **second PCA** is run on the stacked vectors. The resulting parameters c control shape and texture *jointly*: a new image is generated by producing the shape-free texture from c and warping it with the shape that the same c describes.

(Fitting then works by synthesising the model image and minimising its difference to the target, with the parameter-update matrix precomputed from training-set perturbations.)

7. Which of the following statements about atlas-based segmentation practice is **inaccurate**?

- ☐ (A) An atlas consists of an image together with its manual segmentation; it is registered to the unseen patient image and the labels are propagated along the transformation.
- ☐ (B) When an average (mean) atlas is built from a population, the choice of the reference subject does not influence the result — averages built from different references are the same.
- ☐ (C) The quality of an atlas-based segmentation depends on the image similarity between atlas and patient, the presence of pathology, the quality of the registration, and how typical the patient is of the atlas population.
- ☐ (D) Shape-based averaging fuses several segmentations by averaging their distance transforms rather than voting on their labels.

Solution: Answer: B.

This is the **reference-selection bias**: average models built with different reference subjects are demonstrably *not* the same — the lecture shows two different average images from two different references. The bias is precisely what motivates unbiased, group-wise template construction in a natural coordinate frame, rather than anchoring the average to any single individual.

8. Which of the following statements on uncertainty estimation is **incorrect**?

- ☐ (A) One source of uncertainty is that the input x may not uniquely identify the label y .
- ☐ (B) Another source is that the training set D is itself a random sample from the population, and a different draw would have produced different parameters.
- ☐ (C) The Generalized Energy Distance can be evaluated on an ordinary test set with a single annotation per image, which is why it is the most general evaluation of uncertainty.
- ☐ (D) The Expected Calibration Error measures the average gap between the confidence a model assigns and the accuracy it actually achieves at that confidence.

Solution: Answer: C.

The two evaluation routes are exactly reversed. GED (like MMD) is a *distributional* distance between the set of model samples and the set of **observed annotations** — it needs data sets with *multiple labels per image*, such as multi-rater segmentations. When only one annotation per image is available — the most general case — the tool is **calibration**, i.e. ECE: whether the assigned confidence tracks the achieved accuracy.

9. Which of the following statements from the machine-learning recap is **incorrect**?

- ☐ (A) Hyper-parameters — the architecture, the stopping point of the optimization, the learning rate — are chosen using a validation set, ideally as a nested optimization inside the training of the parameters.
- ☐ (B) The training loss is a sum over samples, so the optimal parameters do not have to satisfy any single sample perfectly — they satisfy the samples on average.
- ☐ (C) The prediction error on the test set must be computed with the same differentiable loss that was used for training, since otherwise the numbers are not comparable.
- ☐ (D) For a new sample the prediction is $y \approx f(x; \theta^*)$, and in general $y \neq f(x; \theta^*)$.

Solution: Answer: C.

Backwards on both counts. Differentiability is a requirement of *training* — the loss must be optimized — not of *evaluation*. On the test set one may, and often should, use a different, more application-appropriate function D , precisely in the cases where D is not differentiable (a ranking-based measure is the lecture’s example). The same logic appeared with segmentation: the set-based Dice score is the standard *evaluation* measure even when the network was trained with cross-entropy or soft Dice.

10. “Meaningful perturbation” explains a classifier’s decision by perturbing the input. Which of the following statements describes the method correctly?

- ☐ (A) It searches the training set for the image that most strongly activates the class and presents it as the explanation.
- ☐ (B) It optimizes an information-deletion mask — applying blur, noise or masking — that covers as little of the image as possible while dropping the class score as much as possible.
- ☐ (C) It requires no evaluations of the model, which makes it the cheapest explanation method.
- ☐ (D) It computes the gradient of the class score with respect to the convolutional feature maps and pools it into a heatmap.
- ☐ (E) Because the mask is optimized, the resulting perturbations are guaranteed to remain natural-looking.

Solution: Answer: B.

The key idea is to find the *right* perturbation — a form of information deletion — and study its effect on the prediction: a mask objective balances minimal coverage against maximal

drop in the class score. (D) describes Grad-CAM, a different method. (E) is the method's known caveat, inverted: unconstrained optimization can produce *unnatural* perturbations — adversarial-like artifacts against the explanation — which is why the meaning of the explanation depends on the meaning of the changes applied to the input.

11. A classification network takes grey-scale input images of size 68×68 with one input channel. All convolutions are *valid* (no padding) with stride $S = 1$:

- **Conv1**: 6 kernels with $K = 5 \times 5$ **Pool1**: max pooling 2×2 , $S = 2$
- **Conv2**: 12 kernels with $K = 3 \times 3$ **Pool2**: max pooling 2×2 , $S = 2$
- **Conv3**: 24 kernels with $K = 3 \times 3$

Circle True or False:

- (a) True **FALSE** The output of Conv3 has a spatial size of 15×15 .
- (b) **TRUE** False The global receptive field of a neuron in Conv3 is 20×20 .
- (c) True **FALSE** The global stride of a neuron in Conv3 is 8×8 .
- (d) **TRUE** False The network's convolutional layers have $5 \cdot 5 \cdot 1 \cdot 6 + 3 \cdot 3 \cdot 6 \cdot 12 + 3 \cdot 3 \cdot 12 \cdot 24 = 3\,390$ kernel weights, and including one bias per kernel the total number of parameters is 3 432.

Solution: (a) FALSE — 15×15 is the size *entering* Conv3, not leaving it. Tracking: $68 \xrightarrow{\text{Conv1}} 64 \xrightarrow{\text{Pool1}} 32 \xrightarrow{\text{Conv2}} 30 \xrightarrow{\text{Pool2}} 15 \xrightarrow{\text{Conv3}} 13$. The output is 13×13 .

(b) TRUE — track the receptive field R and jump j from $R = 1, j = 1$ with $R \leftarrow R + (K - 1) \cdot j$ and $j \leftarrow j \cdot S$: Conv1 $\rightarrow R = 5, j = 1$; Pool1 $\rightarrow R = 6, j = 2$; Conv2 $\rightarrow R = 10, j = 2$; Pool2 $\rightarrow R = 12, j = 4$; Conv3 $\rightarrow R = 20, j = 4$.

(c) FALSE — there are only *two* stride-2 poolings, so the global stride is $2 \cdot 2 = 4$, i.e. 4×4 . Strides compound multiplicatively over the layers that actually stride; Conv3 itself has stride 1 and changes nothing.

(d) TRUE — weights: $150 + 648 + 2\,592 = 3\,390$. Biases: one per kernel, i.e. per output channel: $6 + 12 + 24 = 42$, giving 3 432 in total. Pooling layers contribute neither weights nor biases.

12. Which of the following statements are correct (multiple answers are possible)?

- ☐ (A) In FCN-style segmentation, outputs taken at several depths of the network are upsampled by different factors and combined, so that fine detail from shallow layers complements the context of deep layers.
- ☐ (B) A label map may be transformed with bilinear interpolation provided the interpolated values are rounded to the nearest integer label afterwards.
- ☐ (C) Matching the mean and standard deviation of two images, $f(j) = (j - \mu_j) \sigma_i / \sigma_j + \mu_i$, matches second-order statistics and is less sensitive to outliers than min-max normalization.
- ☐ (D) A similarity transformation has 4 degrees of freedom in 2D and 7 in 3D.

- (E) The curl of a transformation field, $\nabla \times T$, quantifies the local volumetric change induced by the transformation.

Solution: Answer: (A), (C) and (D) are correct.

(B) is wrong, and the proposed repair makes it worse rather than better. Bilinear interpolation between labels produces values that are not labels, and rounding them does not restore meaning: between a voxel of label 1 and a voxel of label 3, the interpolated value 2 rounds to label 2 — possibly a completely different structure that is not even adjacent to either. Label maps are transformed with nearest-neighbour interpolation, full stop.

(E) attributes the wrong quantity: the **curl** carries information about the amount of *infinitesimal rotation*. Local volumetric change is quantified by the **Jacobian determinant** ($\det J > 1$ expansion, < 1 compression, $= 1$ none), with the divergence — the trace of the Jacobian — carrying related compression/expansion information.